Meeting 6 of 7 · November 24, 2026

FINAL EXPERIMENTAL FINDINGS

One sealed test: the simple rule held

Sealed held-out evaluation · rule vs learned comparators · reliability–resource trade-offs · limits · publication

The one-time sealed test (Oct 8, 2026) supported all four pre-registered hypotheses: the probe rule was near-Oracle on unseen models, and fixed learned comparators did not reduce material errors. Only four structures were held out.

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Proposed split: Tushar (slides 1–3) · Mayank (slides 4–5) · about 5 minutes



The one-time sealed evaluation

Freeze
rule + 4 learned comparators

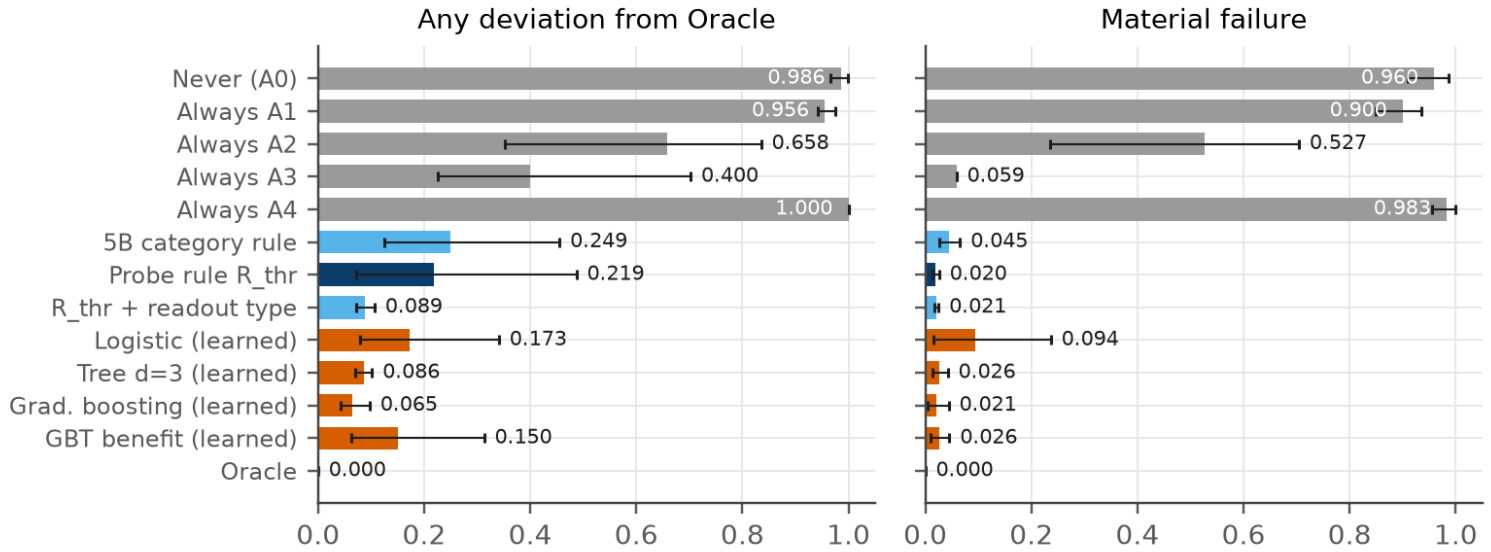
Fingerprint
hash of every frozen decision

Push protocol
hypotheses, metrics, statistics

Unlock once
guard refuses a second unlock

Evaluate
16 unseen models · reads logged

Final held-out evaluation, 5C test (16 unseen models, 4 unseen structures; 95% CIs)



2.0%

material errors, 95% CI 1.4–2.6% (4 held-out structures; likely too narrow)

−0.03% · +0.6%

Brier regret vs Oracle · shots vs Oracle

8.8%

material errors, categorical test (bound: 10%)

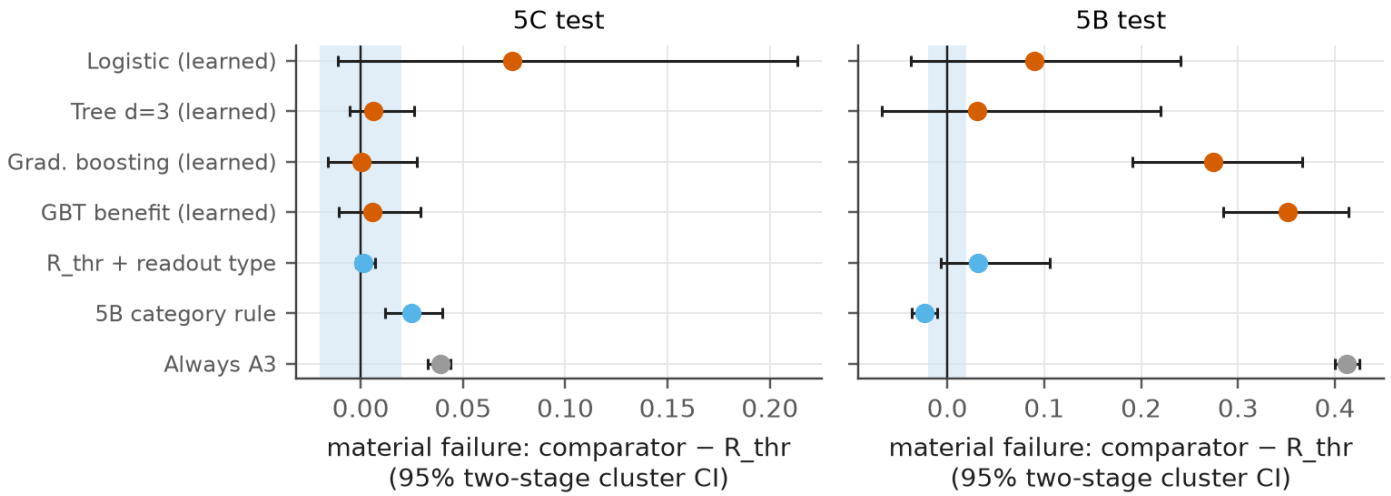
H1 supported **H2 supported** **H3 supported** **H4 supported**

16 unseen models from 4 never-used structures

Source: final_metadata.json, final_policies_5c_test.csv, final_policies_5b_test.csv

The rule versus learned comparators

No learned comparator beats the probe rule on material failure (shaded: ±0.02 margin)



Registered endpoint: material errors

Best learned (gradient boosting) 2.1% vs rule 2.0%; difference +0.0005 [−0.016, +0.027]. 4 structures: intervals likely too narrow.

Any disagreement with the Oracle

Gradient boosting 6.5% vs rule 21.9%, mostly on the unseen mean-Z structure, where the rule's extra choices cost 2 calibration circuits: not material.

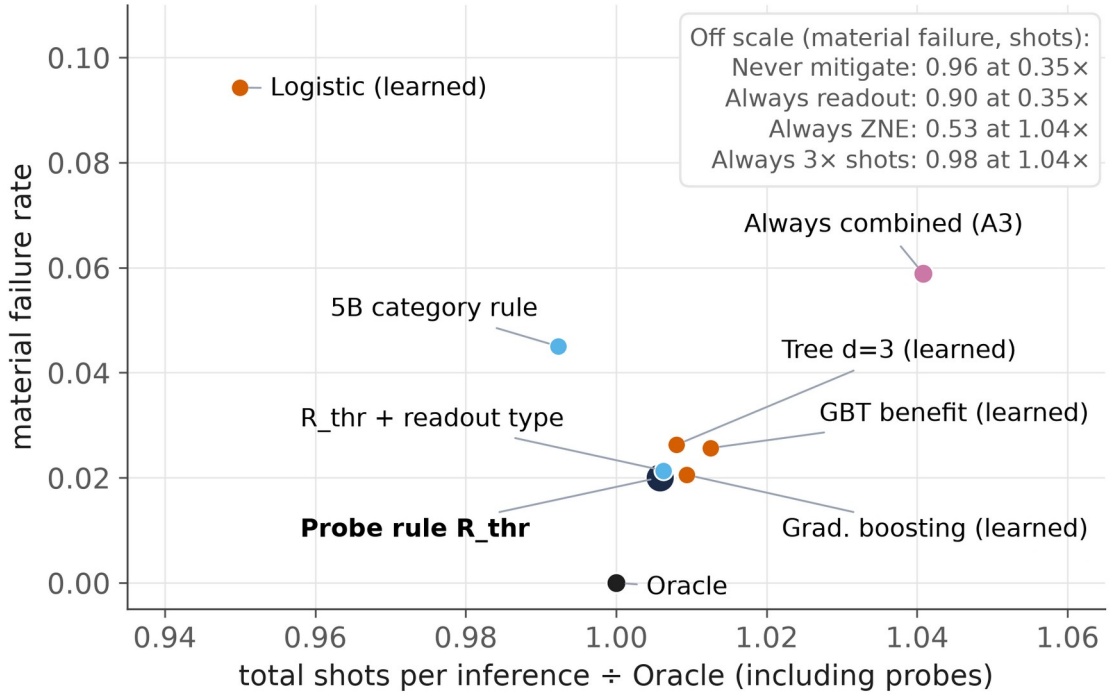
Stability

Gradient boosting: 10.7% material on validation vs 2.1% on test. Rule: 2.0% on both.

Overall, learning did not reduce material errors: fewer on the three parity structures, more on the unseen mean-Z one.

Source: final_paired_vs_rthr_5c_test.csv, final_paired_vs_rthr_5b_test.csv, final_policies_5c_test.csv, p7_policies_5c_val.csv

Reliability, resources, and what the test cannot show



+0.6%

shots vs Oracle

−3.4%

shots vs always-combined

0.35×

shots, never mitigate (96% material)

What the test cannot show

- Always-combined's 5.9% = the 1 gate-free setting in 17: the saving depends on the design
- Only 4 held-out structures: intervals likely too narrow
- H2 is non-superiority on material errors, not proof learning cannot help
- Simulation only · 4 qubits · gate probe nearly blind to thermal noise

Source: final_policies_5c_test.csv, p5c_design.csv, p5c_probe_validity.csv

Contribution and the path to publication

1. Characterization

Noise damaged calibration, not discrimination; mitigation repaired it; extra shots did not.

2. Benchmark

Leakage-safe counterfactual benchmark: 64 VQCs × noise × budget × 5 actions.

3. Pre-registered test

In this setting, a simple probe rule performed competitively with learned comparators.

Status

- Manuscript (LaTeX, 18 pp) and capstone report: complete drafts for review
- References checked against publisher records; team confirmation pending
- Nothing submitted · no acceptance claimed

Recommended route (authors decide)

arXiv
needs an endorser

→

TMLR
claims vs evidence · no fees

→

If needed
IEEE QCE27 or QMI

Pending author decisions: references, author order, venue, arXiv endorser, AI-use disclosure.

In the studied setting, a simple observable probe rule was enough. No quantum advantage is claimed.

Source: deliverables/README.md, deliverables/submission/VENUES.md, deliverables/submission/SUBMISSION_PLAN.md